

MTH105

Examination - January Semester 2026

Fundamentals of Mathematics

Monday, 13 April 2026

04:00 pm - 06:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FOUR (4)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if **FULL** working is shown.
4. This is an **Open-book** examination.

At the end of the examination.

Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

**THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR
SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.**

(Full marks: 100)

Question 1a

Consider the following argument.

$$\begin{aligned}(p \leftrightarrow q) &\leftrightarrow r \\ p &\rightarrow \sim r \\ \therefore p &\rightarrow \sim q\end{aligned}$$

Employ a truth table to determine whether or not the argument is valid.

(10 marks)

Question 1b

Use the Rules of Inference to prove that the following argument form is valid.

$$\begin{aligned}(p \vee q) &\vee r \\ r &\rightarrow p \\ q &\rightarrow (p \vee r) \\ \therefore p\end{aligned}$$

(10 marks)

Question 2a

Give a counter-example to show that the following statement is false:

$$\forall x \in \mathbb{N}, \forall y \in \mathbb{N}, \forall z \in \mathbb{N}, \forall w \in \mathbb{N}, (y = zw \wedge x \text{ divides } y) \rightarrow (x \text{ divides } z \vee x \text{ divides } w)$$

(5 marks)

Question 2b

Provide the negation of the following statement:

$$\forall a \in \mathbb{N}, \forall b \in \mathbb{N}, ((b \text{ divides } a^{b-1} - 1) \wedge a > 1 \wedge a < b) \rightarrow (b \text{ is prime})$$

(5 marks)

Question 2c

Consider the sequence of positive integers defined by $x_1 = 1, x_2 = 2$ and for $n \geq 2$, x_{n+1} is defined as the least integer p , if such a p exists, such that $p > x_n$ and there is exactly one pair (i, j) with $1 \leq i < j \leq n$ such that $p = x_i + x_j$ (that is, p is expressible uniquely as a sum $x_i + x_j$ of two earlier distinct terms x_i and x_j). Employ a proof by contradiction to show that the set $\{x_n \mid n \in \mathbb{N}\}$ is infinite.

(10 marks)

Question 3a

Employ either a proof by contraposition or a proof by contradiction to show the following statement:

Suppose $a \in \mathbb{R}$. If $3a^2 - 2a + 4$ is irrational, then a is irrational.

(10 marks)

Question 3b

Implement the Euclidean Algorithm to find $d = \gcd(2028, 888)$ and express d as an integral linear combination of 2028 and 888.

(10 marks)

Question 4a

Use mathematical induction to show that

$$\sum_{r=1}^n \frac{1}{(3r+1)(3r-2)} = \frac{n}{3n+1}$$

for all positive integers n .

(15 marks)

Question 4b

Let $A = \mathbb{Z} \times \mathbb{Z}$. Define a relation R on A by the following rule: For every $(a, b) \in A$ and every $(c, d) \in A$,

$$(a, b) R (c, d) \Leftrightarrow |a - b| = |c - d|.$$

Determine whether R is an equivalence relation on A . If R is an equivalence relation on A , describe the equivalence classes of R .

(10 marks)

Question 5

Let X be a non-empty set and fix any element $x \in X$. Define a function $f : \mathcal{P}(X) \longrightarrow \mathcal{P}(X)$ by

$$f(A) = A \triangle \{x\}.$$

(Here $\mathcal{P}(X)$ denotes the power set of X and $A \triangle B$ denotes the symmetric difference of sets A and B .)

Question 5a

Determine explicitly the sets $f(\emptyset)$ and $f(X)$.

(5 marks)

Question 5b

Determine with justification whether f is injective and/or surjective for any nonempty set X and any $x \in X$.

(10 marks)

----- END OF PAPER -----